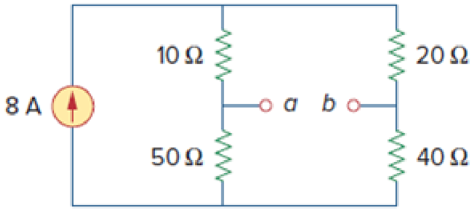
Chapter 4, Problem 59P

Problem

Determine the Thevenin and Norton equivalents at terminals *a*-*b* of the circuit in Fig. 4.125.

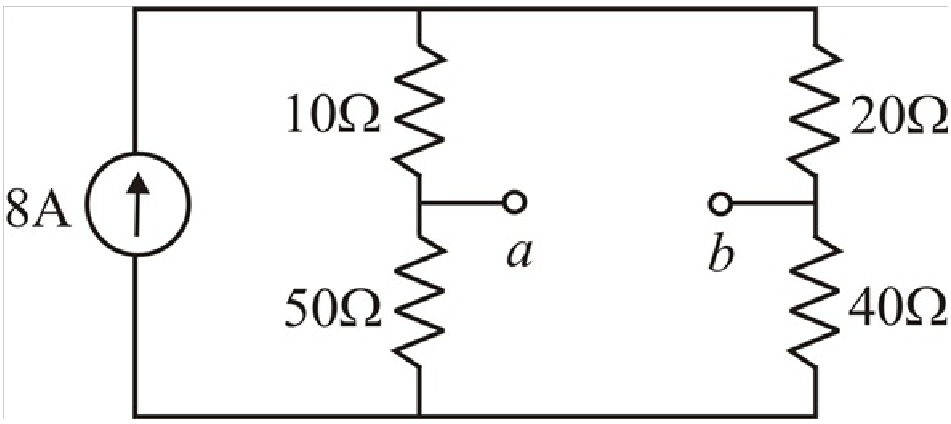
Figure. 4.125:



Step-by-step solution

Step 1 of 13

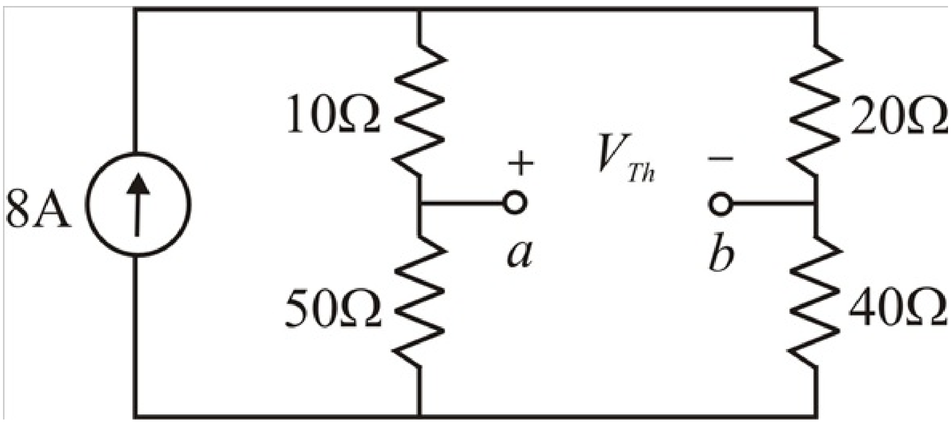
Given circuit diagram



Step 2 of 13

Thevenin's Equivalent Circuit:

To determine Thevenin's equivalent voltage (V_{Th}) between a and b terminals consider below circuit



Step 3 of 13

From the above circuit we can write

$$V_{Th} = V_a - V_b$$

By using current division rule current flowing through 50Ω resistor is

$$I_{50\Omega} = 8\left(\frac{20+40}{20+40+10+50}\right)$$

$$I_{50\Omega} = 8\left(\frac{60}{120}\right)$$

$$I_{50\Omega} = 4\text{A}$$

From the above circuit

$$V_a = I_{50\Omega}(50)$$

$$V_a = (4)(50)$$

$$V_a = 200\text{V}$$

Step 4 of 13

Current flowing through 40Ω resistor is

$$I_{40\Omega} = 8 - I_{50\Omega}$$

$$I_{40\Omega} = 8 - 4$$

$$I_{40\Omega} = 4\text{A}$$

From the above circuit

$$V_b = I_{40\Omega}(40)$$

$$V_b = (4)(40)$$

$$V_b = 160\text{V}$$

$$V_{Th} = 200 - 160$$

$$\boxed{V_{Th} = 40\text{V}}$$

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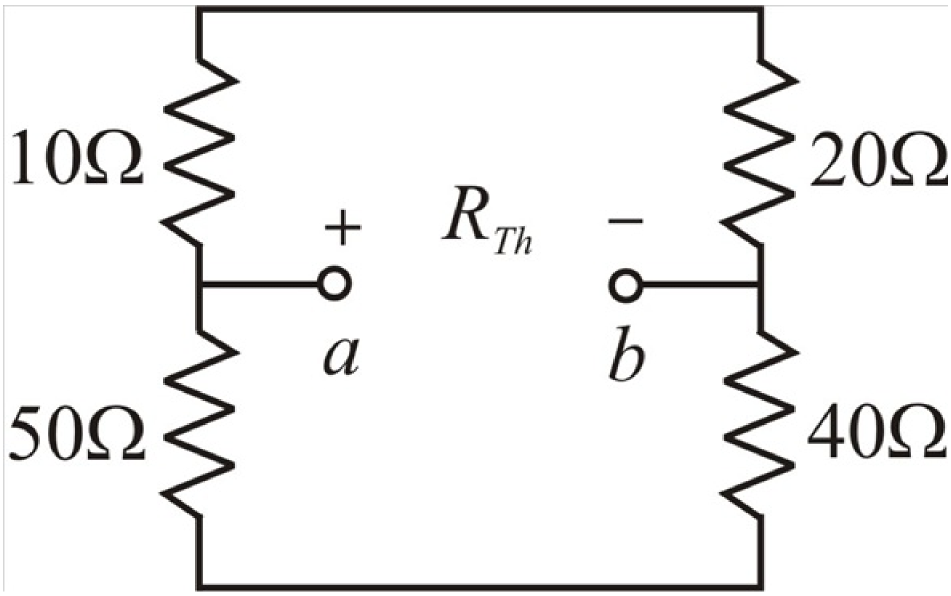
To determine Thevenin's resistance (R_{Th}) between a and b terminals consider below circuit



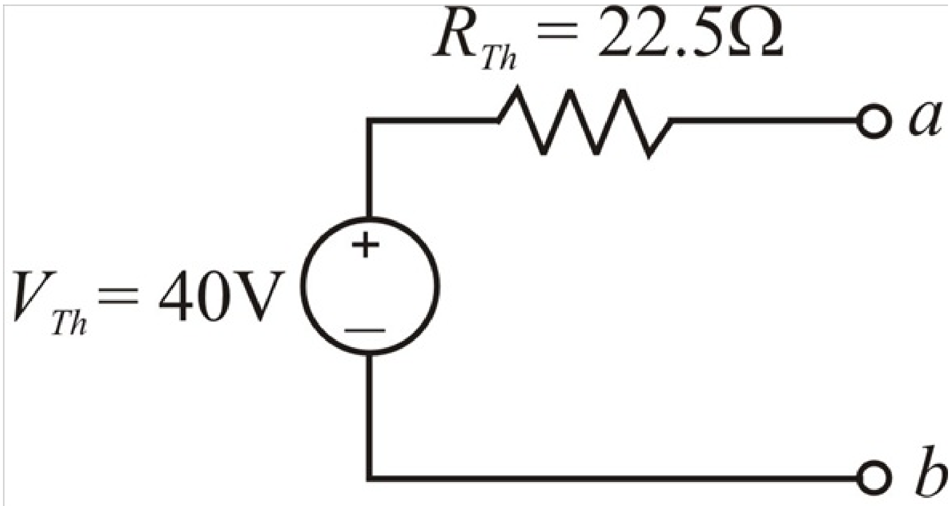
$R_{Th} = (10+20) \parallel (50+40)$

$R_{Th} = 30 \parallel 90$

$R_{Th} = \dfrac{30 \times 90}{30+90}$

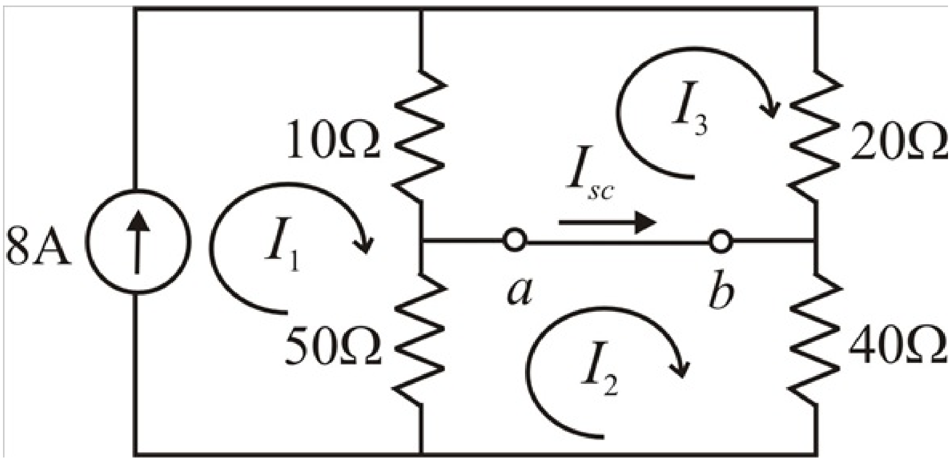
$\boxed{R_{Th} = 22.5\Omega}$

Therefore Thevenin's equivalent circuit between a and b terminals



Norton's Equivalent Circuit:

To determine Norton's equivalent current (I_N) between a and b terminals consider below circuit



Here $I_N = I_{sc}$ (short circuit current)

From the above circuit

$$I_1 = 8\text{A}$$

$$I_{sc} = I_N = I_2 - I_3$$

Applying KVL to loop 2 gives

$$50(I_2 - I_1) + 40(I_2) = 0$$

$$-50I_1 + 90I_2 = 0$$

$$-50(8) + 90I_2 = 0$$

$$90I_2 = 400$$

$$I_2 = 4.4444\text{A}$$

Step 10 of 13

Applying KVL to loop 3 gives

$$10(I_3 - I_1) + 20(I_3) = 0$$

$$-10I_1 + 30I_3 = 0$$

$$-10(8) + 30I_3 = 0$$

$$30I_3 = 80$$

$$I_3 = 2.6667\text{A}$$

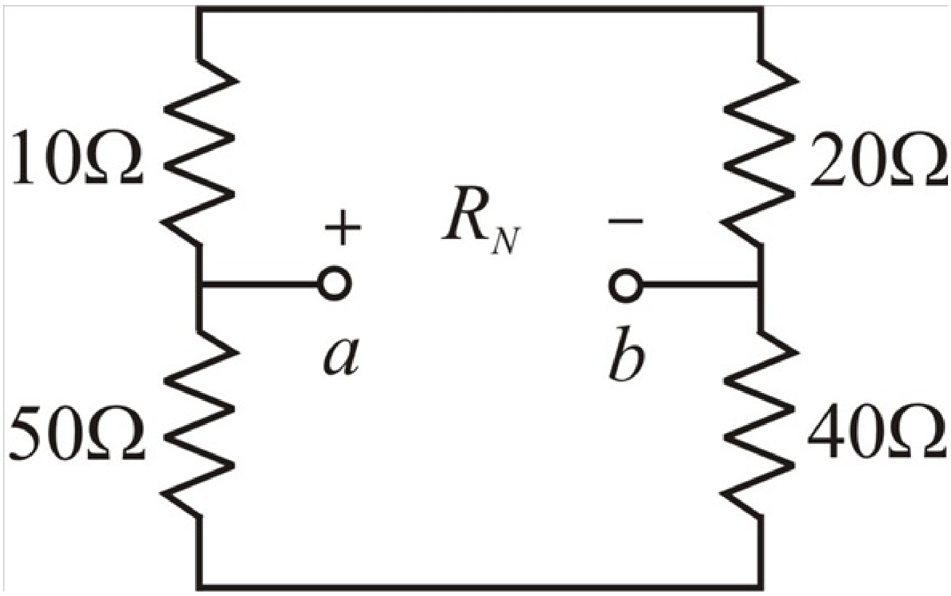
$$I_N = 4.4444 - 2.6667$$

$$I_N = 1.7777\text{A}$$

$$\boxed{I_N = 1.78\text{A}}$$

Step 11 of 13

To determine Norton's resistance (R_N) between a and b terminals consider below circuit



Step 12 of 13

$$R_N = (10+20) \| (50+40)$$

$$R_N = 30 \| 90$$

$$R_N = \frac{30 \times 90}{30+90}$$

$$\boxed{R_N = 22.5\Omega}$$

Step 13 of 13

Therefore Norton’s equivalent circuit between a and b terminals